RAJALAKSHMI ENGINEERING COLLEGE

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Y2

Started on	Wednesday, 30 September 2026, 9:37 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 9:38 AM
Time taken	49 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Binary search function to find the first element strictly greater than x
int upper_bound(int tails[], int len, int x) {
    int low = 0;
    int high = len;

    while (low < high) {
        int mid = low + (high - low) / 2;
        if (tails[mid] <= x) {
            low = mid + 1;
        } else {
            high = mid;
        }
    }
    return low;
}

int main() {
    int n;

    // Read the size of the sequence
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }

    int *arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL) {
        return 1;
    }

    // Read array elements (skips any accidental non-numeric characters like commas/brackets)
    for (int i = 0; i < n; i++) {
        // Tolerates competitive programming formats with brackets/commas
        while (scanf("%d", &arr[i]) != 1) {
            scanf("%*c");
        }
    }

    // tails[i] stores the smallest tail of all non-decreasing subsequences of length i+1
    int *tails = (int *)malloc(n * sizeof(int));
    if (tails == NULL) {
        free(arr);
        return 1;
    }

    int len = 0; // Length of longest non-decreasing subsequence

    for (int i = 0; i < n; i++) {
        // Find the index of the first element strictly greater than arr[i]
        int idx = upper_bound(tails, len, arr[i]);
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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